



Simulations for Technology Evaluation

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Abstract

Denmark aims to significantly increase its production of renewable electricity in the coming years. In practice, this is not straightforward, as the electricity grid is already under pressure and grid expansions take time. One way to deal with this challenge is through Energy Communities, where households produce electricity locally and share it with each other instead of relying entirely on the central grid. In this project, the focus is therefore on investigating whether Energy communities can reduce electricity costs and CO₂ emissions compared to a situation where households only use grid electricity.

To study this, an agent-based simulation model was built in AnyLogic. The model represents an energy community consisting of 137 households located in the DK1 electricity region. Each household is modelled individually using realistic hourly electricity demand data. Local electricity production is included through rooftop photovoltaic panels and, in some scenarios, a community-owned wind turbine. The model also incorporates peer-to-peer electricity trading, grid interaction, investment costs and CO₂ emissions. Six scenarios are analysed, ranging from full reliance on the electricity grid to advanced Energy Community configurations with widespread PV adoption, peer-to-peer trading and a community wind turbine. The scenarios represent different stages of technology adoption and allow comparison of electricity costs, emissions, and self-sufficiency.

The results showed that introducing local renewable generation and energy sharing can lower both electricity costs and CO₂ emissions at the community level. It also showed that allowing peer-to-peer trading leads to a reduction in total electricity costs compared to the baseline case where there are no sharing. Scenarios with higher shares of renewable generation, especially when combined with peer-to-peer trading and a community wind turbine, show even larger cost reductions and, in some cases, result in net-negative CO₂ emissions over the year.

The sensitivity analysis shows that peer-to-peer trading also makes the system less sensitive to changes in renewable production, as locally produced electricity can be used more effectively within the community. This study's results suggest that Energy Communities can reduce reliance on the grid, lower emissions, and improve economic performance while increasing self-sufficiency.

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Abbreviations List

DK	Denmark
CO ₂	Carbon dioxide
PV	Photovoltaic
RE(S)	Renewable Energy (Sources)
WT & WP	Wind Turbine & Wind Power
P2P	Peer-to-peer
UML	Unified Modelling Language

1. Introduction

Denmark is split into two electricity regions since 2012; The West Denmark grid area (DK1) covers Jutland, Fyn Island and the remaining islands to the west of the Great Belt, while the East Denmark grid (DK2) covers Zealand Island. A map illustrating the two grids and their connections outside of Denmark (DK1 with Germany and DK2 with Sweden), is shown in Figure 1 (Saez-Gallego, 2016). In this report, the focus is on the former grid, and specifically in a community with a 400 kVA transformer.

Denmark has set a very ambitious target, which is to quadruple the 2022 electricity production from solar and wind by 2030 (Danish Council on Climate Change, 2024). There have been several issues regarding this goal, with the most prominent being its slow grid development. Experts have pointed out that Energy Communities may be the answer to most of these issues, as they present an alternative solution to using the central power grid. In principle, Energy Communities utilize renewable energy that is produced geographically close to them, which makes electricity overall cheaper (Sokołowski, 2023). The integration of local renewable energy would also result in lower CO₂ emissions, as the focus is on 'green' energy production, and possibly in 'energy justice'. Energy justice is a theoretical term which is based on the principle of all citizens having access to affordable, sustainable and reliable energy (McCauley, 2019).

The Danish electricity transmission system ultimo 2012

Legend

- 150 kV- or 132 kV-substation
- 220 kV-substation
- 400 kV-substation
- Power station
- ⋈ Offshore wind farm

- 150 kV- or 132 kV-cable
- 150 kV- or 132 kV-overheadline
- 220 kV-cable
- 220 kV-overheadline
- 400 kV-cable
- 400 kV-overheadline
- HVDC cable
- HVDC overheadline

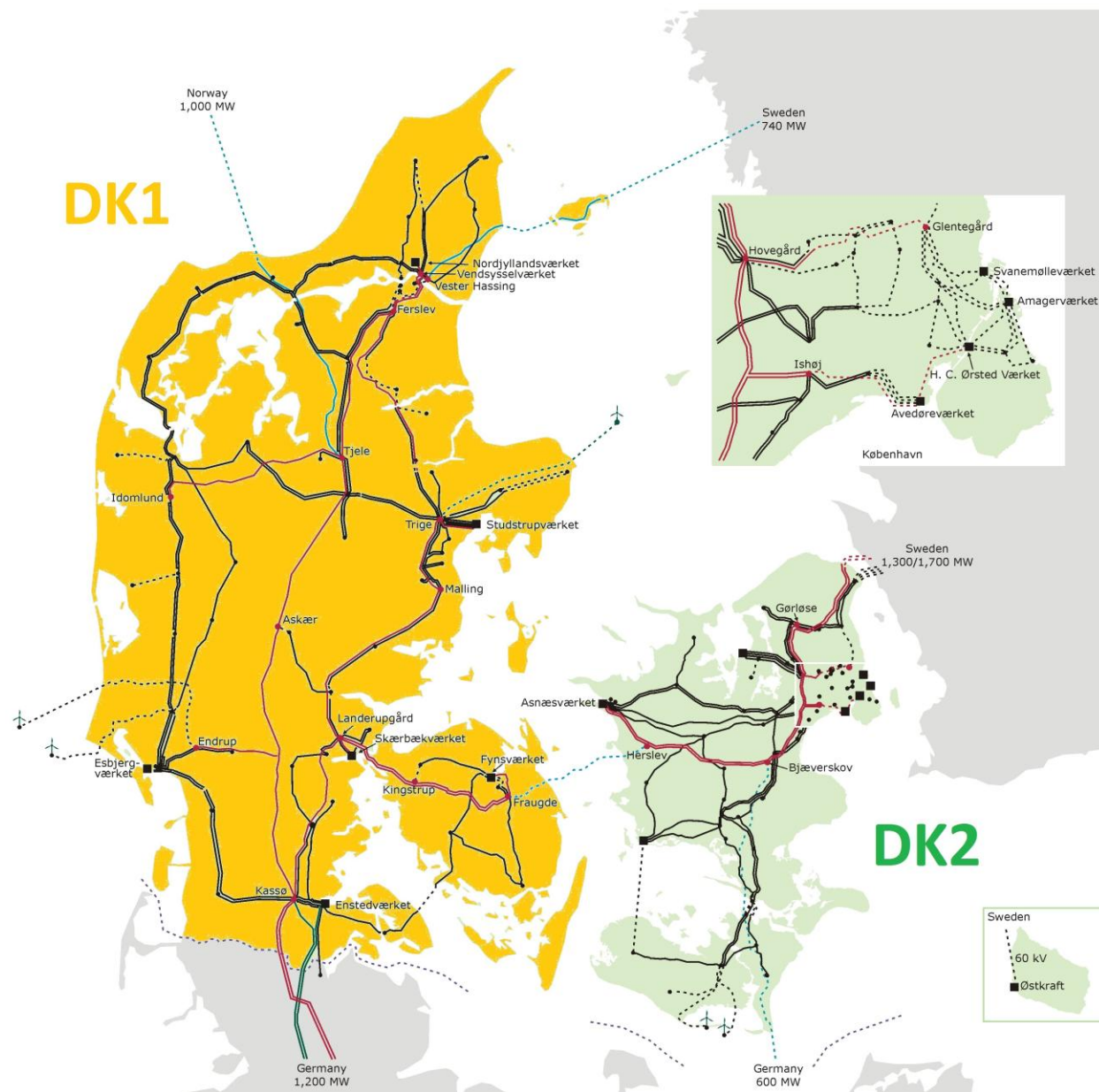


Figure 1: Map of Denmark's two electricity grids. DK1 is highlighted in yellow, and DK2 is highlighted in a pale green colour (Saez-Gallego, 2016).

The aim of this report is to prove this hypothesis regarding economic and environmental factors. In case electricity in Energy Communities is both cheaper and less environmentally burdening, then energy justice will also have been achieved. Therefore, the research question is essentially: “Are the Energy Communities more financially beneficial for the residents and more environmentally friendly than normal Danish communities?”

This research question, while vague on its own, is paired with detailed research objectives:

1. Electricity Consumption and Production Patterns of a community.
 - a. In order to provide a robust answer to the research question through a simulation, the simulation needs to be based on an accurate model of a community. In this case, it is essential to simulate the electricity consumption behaviors of the residents of a community of 137 households.
 - b. As far as production is concerned, the research objective refers to the PV panels, and potential other technologies (e.g. Wind Turbine) that may be included in an Energy Community. These technologies are susceptible to weather changes; a sunny day means more energy produced by PV panels and so on. It is essential to include weather and seasonal variations, as well as the time of day. These exogenous factors are considered in the simulation.
2. Electricity Sharing.
 - a. As mentioned, Energy Communities rely heavily on peer-to-peer electricity trading, where the households that produce excess electricity first share it with their peers and secondly, in case there is still excess electricity after the community’s demand has been met, is then sold to the main energy supplier. This objective encloses the development and testing of the peer-to-peer electricity sharing strategy with several scenarios, including the potential economic and environmental benefits.
3. Renewable Energy Resources (RES) Integration.
 - a. Energy Communities are based on the incorporation of RES as a communal resource for their energy supply. Even if the PV panels are installed in individual households, the fact that they may funnel the surplus electricity to their peers and then to the main grid, is enhancing the communal feeling. In order to simulate the RES

Integration of existing Energy Communities in the model community of this report, the simulation includes the deployment of solar panels and a small-scale communal Wind Turbine. In all scenarios assessed in this report, the financial factors, such as capital & installation costs, are considered. Also, the simulation is also used to assess the impact of RES on community-level energy and self-sufficiency.

4. Resilience of the Energy Community

- a. As the community is reliant on RES, it is important to check how much resilience there is to disturbance. In order to do that, the sensitivity of each scenario is tested in regards of one factor, in this case, the PV electricity production. Since the community, in all scenarios, is always connected to the main grid, in case of low to no RE production, the electricity demand of all households will be met by the main electricity supplier. The community, under no circumstances, will be left without continuous power supply, but the question of resilience remains.

In the above breakdown of the research objectives, the different technologies that will be tested have been mentioned. Nevertheless, it is important to properly introduce them, as they are the key to the simulation and the community. Firstly, there is the main electricity supplier, to which the households are connected. Next, there are the PV panels (also called solar panels), which are installed on the roofs of individual households and connected to them. A technology that affects the electricity distribution in the community is the P2P electricity trading; in the case of P2P electricity trading, the households are interconnected, in top of their grid connection, and the surplus electricity of the individual PV panels is funnelled to the peers. In the case the excess solar power covers the community's demand (hourly) and there is surplus left, the latter gets sold to the main supplier through the main electricity grid. There are scenarios where there is no P2P electricity trading, which is why it is important to mention it as a cutting-edge technology, as, globally, its implementation remains limited (Kamini Singh, 2025). Wind Turbines (WT) are next on the list, after P2P electricity trading. It is a very common RES, widely deployed in Denmark, but in the case of Energy Communities, the WT used are usually smaller than the offshore ones and with lower capacity (Kamini Singh, 2025).

The ecosystem transition stages for Energy Communities are three, and they each represent a different adoption of the state-of-the-art technology:

1. *Incumbent Stage*. In this report this stage is called the 'incumbent' stage, where no technology is adopted other than grid connection. All households in question rely on the main electricity grid for their electricity needs.
2. *Niche Technology Introduction Stage*. This stage follows the 'incumbent' one, and it represents the start of the adoption of new technologies. In this case, the new technologies are PV panels, anything other than the main grid. The difference with the following stage is the interconnection with the households; in the Niche Technology Introduction stage the households are connected to the grid. If the households have individual PV panels installed, then they are benefitting solely from them, and they do not share their surplus electricity with their peers; they sell it back to the main supplier through the electricity grid.
3. *Niche Technology Adoption Stage*. This is the stage of state-of-the-art technologies; in this case, state-of-the-art technologies are the peer-to-peer electricity trading and the existence of a community Wind Turbine. This is the only stage where the community, which is the subject of this report, can be considered an 'Energy Community'. While WT on their own would be categorised with solar panels in the second stage, the community WT allows for the 137 households to be considered an Energy Community; therefore, in the context of this report, it is included in the last stage.

While the ecosystem does not present with a lot of challenges, there are some pain points. The most prominent challenge is the intermittency of RES. In both cases of RES (solar and wind power), the technologies are reliant on the weather; hence, there are periods of no RE production, while the demand for electricity remains stable. On the other hand, there could be cases where the electricity production from RES is higher than the demand at this hour, resulting in surplus electricity for the community. Since in the Energy Community of this report there is no electricity storage, the consumption-production ratio is balanced through the main grid. This is also a challenge in some cases, as it affects revenues, self-consumption and emissions. Lastly, there is the challenge of justice. As mentioned earlier, one of the main goals of an Energy Community is

that of 'energy justice', but there are some cases where it could be challenging. A community-owned WT introduces collective investment and shared revenues, while privately owned PV creates uneven benefits. The allocation rules (who pays, who benefits) influence equity and uptake. This issue raises questions like "Who can afford PV or participate in collective ownership?" and "Does P2P or WT revenue benefit all households equally?".

Overall, the scenarios developed for the present report were built with the research question, research objectives, state-of-the-art technologies and challenges to mind, and they ought to provide adequate answers to all questions and situations.

2. Methodology

2.1 Methodology Introduction

The model system in this report is based on Energy Communities in Denmark. Energy communities are local groups of citizens, authorities and companies that collectively own and manage renewable energy production (e.g. PV panels, wind turbines). In this report, the focus is on an Energy Community of 137 electricity consumers (assuming households) in DK1.

This report employs agent-based modelling to simulate interactions among entities within the targeted ecosystem. This section will identify the chosen simulation methodology, discussing its advantages and limitations.

Modelling of systems is categorised in three types as seen in Figure 1 (Grigoryev, 2020), discrete event modelling, agent-based modelling, and system dynamics. The three models have varying levels of abstraction.

The system dynamic method focuses on grouped quantities and feedback loops in systems. Its high abstraction level is abstracting individual objects into stocks and flows. The System dynamic method is usually utilised to understand the systems' dynamic behaviour. It involves abstract concepts such as morale awareness, knowledge, and the impact of advertising. These abstract concepts may have something other than concrete counterparts in the real world, which makes the precision of the model complex to ensure in real-world scenarios. In addition, system

dynamics does not consider events of the individual interest, like purchase decisions. Events are aggregated into flows. In some scenarios, the individual decisions are critical and require a high level of detail, so the system dynamic in such scenarios is lacking.

Discrete event modelling uses system processes, entities, and resources to model sequential steps. In this method, the entities move within processes and interact with system resources. This model is suitable for analysing systems where agents' discrete interactions are centralised. The discrete event modelling has a low abstraction level compared with system dynamics. The models are stochastic; this stochasticity is due to probability distribution for service times and agent arrivals. Therefore, the model outcomes/results are inherently uncertain and require a relatively long running time or some replications.

Agent-based modelling is usually used to model systems containing active objects such as people, factories, business units, and products. The active objects can be translated into agents that operate and interact in pre-established environments. From this environment, agents can gather the necessary information to pursue their own strategy and make decisions, but their overall knowledge of the system's state is limited. In this model, agents are autonomous and behave differently but follow a predefined rule.

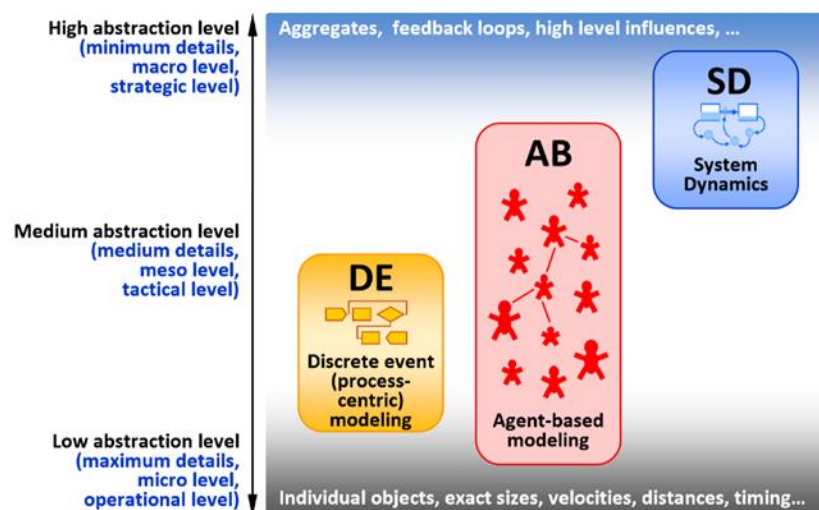


Figure 2: Categories of system modelling tools and levels of abstraction (Grigoryev, 2020).

The abstraction level in the agent-based model is flexible, as it can range from high to low. This flexibility allows for modelling agents with different levels of detail.

The chosen simulation method for this report is agent-based modelling, which is computationally efficient due to localized decision-making at the agent level. This approach provides a detailed understanding of the ecosystem by simulating real-life scenarios. In this paradigm, actors and objects within the system are represented as agents, with their behaviours, interactions, and data flows defined using a programming language. This allows the model to illustrate interdependencies and dynamics within the system. However, agent-based modelling has limitations, including challenges in obtaining necessary data and difficulties in validating results.

2.2 Simulation architecture

This section introduces each agent, explains what it represents in the local energy system, and summarises its function within the simulation.

The simulation model is composed of several interacting agents, each representing a key component of the energy community. These agents exchange electricity flows, economic information, and operational signals hourly.

The implemented energy community model consists of six main agent types. Each agent represents a specific entity within the community and carries out a specific role in the simulation. Table 1 summarises the agents, their interfaces, and their main functions.

Table 1: Agents, interfaces, and main functions of the simulation

Ecosystem actor/object	Represents	Model interface name	Model functionality
Consumer	A domestic electricity user (one household).	I_Consumer	Interacts with its Meter, receives locally produced energy as PV and wind, participates in P2P exchanges, receives bills from supplier, and calculates its (lost) revenues, PV investment costs, and CO ₂ emissions.

Electricity supplier (Supplier)	The external electricity retailer and grid connection point.	I_Supplier	Provides hourly electricity prices and CO2 intensity, settles imported and exported electricity, keeps track of and sends out customer bills, and determines the cost associated with grid purchases and export compensation.
Meter	The household's smart meter.	I_Meter	Measures the household's hourly electricity balance. Splits the energy flows into contributions from PV, wind, P2P exchanges, and the supplier. These measurements form the basis for household-level billing and emissions calculations.
PV (IndividualSolar Panel)	The rooftop photovoltaic system owned by a household.	I_IndividualSolarPanel	Computes household-specific PV generation based on installed capacity and solar-efficiency coefficients and computes the hourly investment cost of the PV system.
WindTurbine	A shared, community-owned wind turbine serving all 137 households.	I_WindTurbine	Calculates hourly wind power output using a wind coefficient and fixed turbine capacity, and supplies this energy to the Community agent. Also computes the hourly investment cost of the wind turbine.
Community	The coordinator of household energy flows and the local P2P market.	I_Community	Aggregates net loads across households, classifies buyers and sellers, allocates available PV and wind energy, administers P2P exchange, settles remaining wind energy flows with the Supplier, and calculates self-sufficiency index of the community.

In the real Danish electricity system, the responsibilities represented by the agents in this model are divided among several actors. Smart meters are operated by Energinet's DataHub, which collects and distributes consumption data to suppliers and grid operators. Also, PV and wind output are typically measured through separate monitoring equipment, and DSOs handle grid constraints and local energy balancing. To maintain clarity and focus, the model combines these real-world responsibilities into simplified agents, FX: a single Meter, a single Supplier. This

abstraction allows isolate the household-level interactions, renewable integration, and P2P mechanisms that are central to the analysis.

2.3 Agent Hierarchy & Agent Logic

Figure 3 illustrates the agent hierarchy used in the energy community model. The hierarchy is not fully flat: the top-level Energy Community agent serves as a container for internal components and includes both the shared Wind Turbine agent and a population of 137 Consumer agents, each representing a household in the community.

Each Consumer agent contains exactly one Meter and one PV agent. The Meter functions as the household's smart meter and records hourly electricity flows, while the PV agent represents the rooftop photovoltaic system and computes household-specific solar production and its associated investment costs.

The Supplier agent is positioned outside the Energy Community and represents the external electricity retailer and grid connection point. It provides hourly price and CO₂ intensity signals and settles any remaining imports or exports once internal PV, wind, and P2P exchanges have been processed.

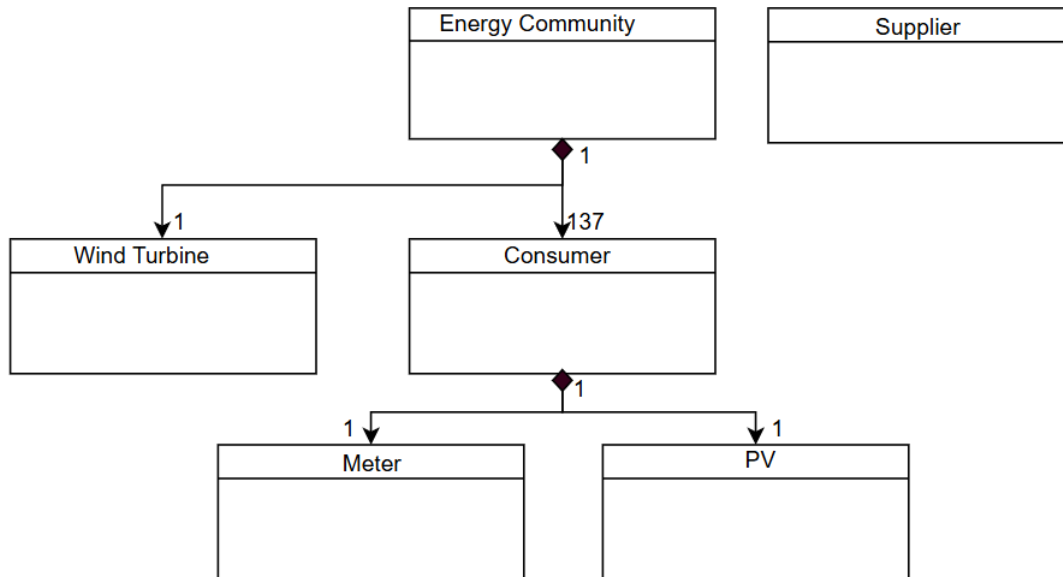


Figure 3: Agent hierarchy

It should be noted that this hierarchy represents the general model structure. Not all agents are active in every scenario—for example, Scenario 1 has no PV systems, and Scenarios 1–3 and base case do not include the wind turbine. The diagram, therefore, reflects the full model architecture, rather than the configuration of any single scenario.

At the beginning of each hour, input information is collected. Hourly electricity price and CO₂ intensity for the DK1 region, weather-based coefficients that determine the solar and wind generation potential for that hour, and the electricity demand of each of the 137 households. In scenarios without PV or wind, the corresponding generation inputs are simply not used. Once information inputs are available, the EnergyCommunityLogic evaluates the energy balance of the community. At the beginning, each household’s net load is calculated by subtracting its PV production (if present) from its demand. Based on this net load in the P2Pto solar scenarios, households are divided into two groups: buyers, who require electricity, and sellers, who have surplus PV energy available. These quantities are stored internally so they can be matched later. In scenarios with a wind turbine, the wind generation is distributed among households proportionally to their remaining demand, so that households with higher demand receive a

larger share of the available wind energy. Any demand that is still unmet after wind allocation is supplied by the grid, while surplus wind generation (if any) may be exported depending on the scenario assumptions.

In solar + P2P scenarios, households are first divided into buyers and sellers based on the electricity remaining after self-consumption. Households with excess PV production act as sellers, while households with unmet demand act as buyers. The available surplus PV is then shared within the community through P2P trading. The buyers' electricity share is proportional to their demand, they need to be met, meaning that households with higher demand receive a larger share of the available PV electricity. If the total PV surplus is not sufficient to meet the community's demand, the needed electricity is imported from the grid. On the other hand, if any surplus production from PV remains after all buyers have been supplied, it is exported to the grid in scenarios where export is allowed.

After the internal matching is complete, the community settles with the electricity supplier. At this point, if any electricity demand that has not been covered locally, is supplied by the grid, and surplus renewable production can be exported in scenarios where this is allowed. This results in a final hourly electricity balance for each household.

These energy flows are stored in each household's meter, and the associated consumer updates its electricity bill, any P2P or export revenues, investment costs and CO₂ emissions for that hour. The community agent then aggregates these values across all households to produce the community-level indicators used in the scenario comparison.

This sequence repeats for every hour of the simulated year. Figure 4 illustrates the main decision-making process that unfolds during each operating hour of the simulation. It should be noted that attributions and methods shown in the UML below do not represent all variables and functions implemented in the model.

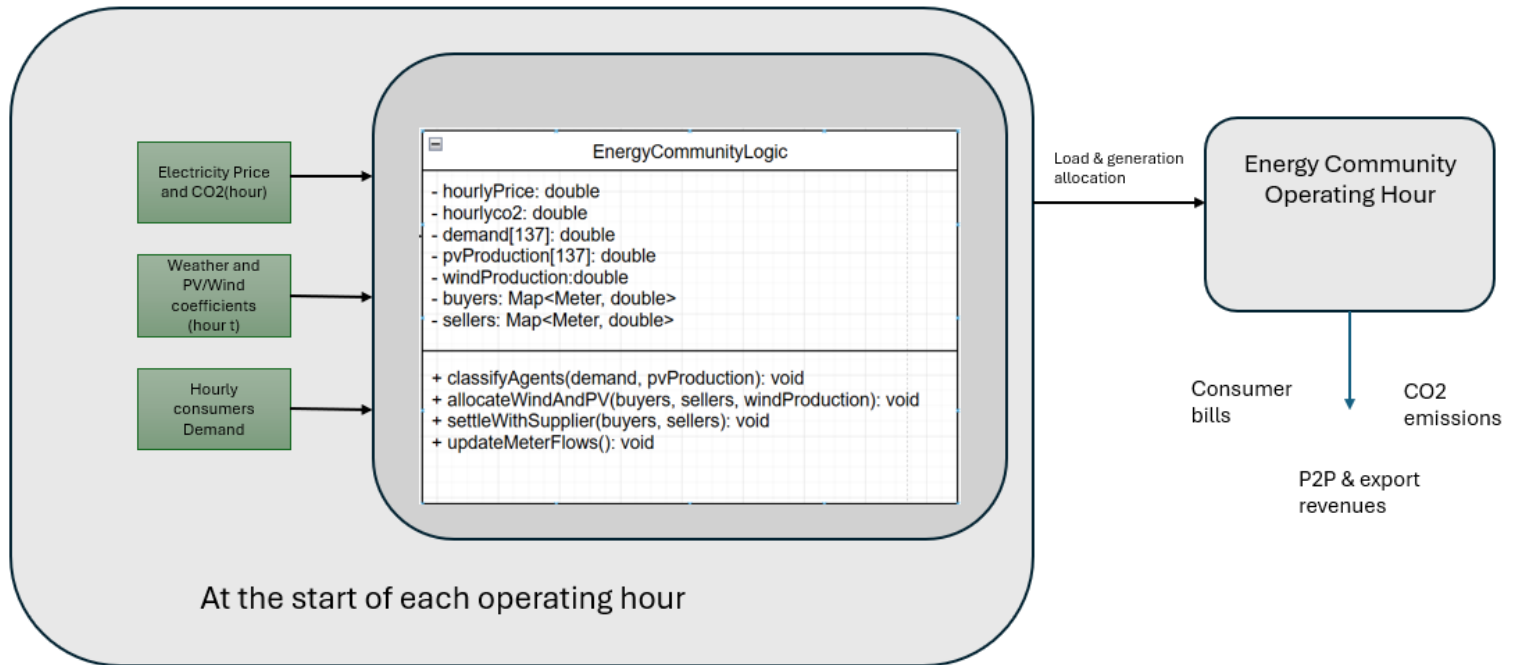


Figure 4: Implementation logic of the EnergyCommunityLogic component

2.4 UML Sequence Diagrams

In order to better showcase the interactions between the agents and objects of the simulations, UML (Unified Modelling Language) Sequence Diagrams were created for the scenarios. A UML Sequence Diagram is a visual representation of how the objects of a system interact with each other over time, showing the order of messages exchanged between them. The messages sent from one agent to another are represented by horizontal straight arrows, with the point of the arrow turned to the receiver. In the diagrams below, there are also 'self-messages', also called self-calls, and instead of straight arrows, they are rectangular loops. In the cases of self-calls, the object is calling its own operation. UML Sequence Diagrams are a great illustration of the sequence of processes; while they do not showcase the duration of a process, the phases of the simulation are in order of progression on the vertical axis. As mentioned above, the base case and scenarios have certain similarities, such as the inclusion of a wind turbine (scenarios 4 and 5) and the peer-to-peer electricity sharing (scenarios 2,3,4,5). Below, there are four UML Sequence Diagrams, one for the base case, one for Scenario 1, one for Scenarios 2 and 3, and one for Scenarios 4 and 5. The difference between Scenarios 2 and 3, which lies in the percentage of PV

panel adoption in the Energy Community, is in the numbers, and therefore, the two UML Sequence Diagrams do not differ from each other, as all communications and interactions, as well as the actors and objects, remain the same. Therefore, there is one UML Sequence Diagram corresponding to both these scenarios. Similarly, there is one UML Sequence Diagram corresponding to Scenarios 4 and 5, as their differences are also numerical and not representative in the UML Sequence Diagrams. Nevertheless, in order to showcase the similarity between the aforementioned scenarios, the remaining UML Sequence Diagram for Scenarios 3 and 5, not inserted below, can be found in Appendix B.

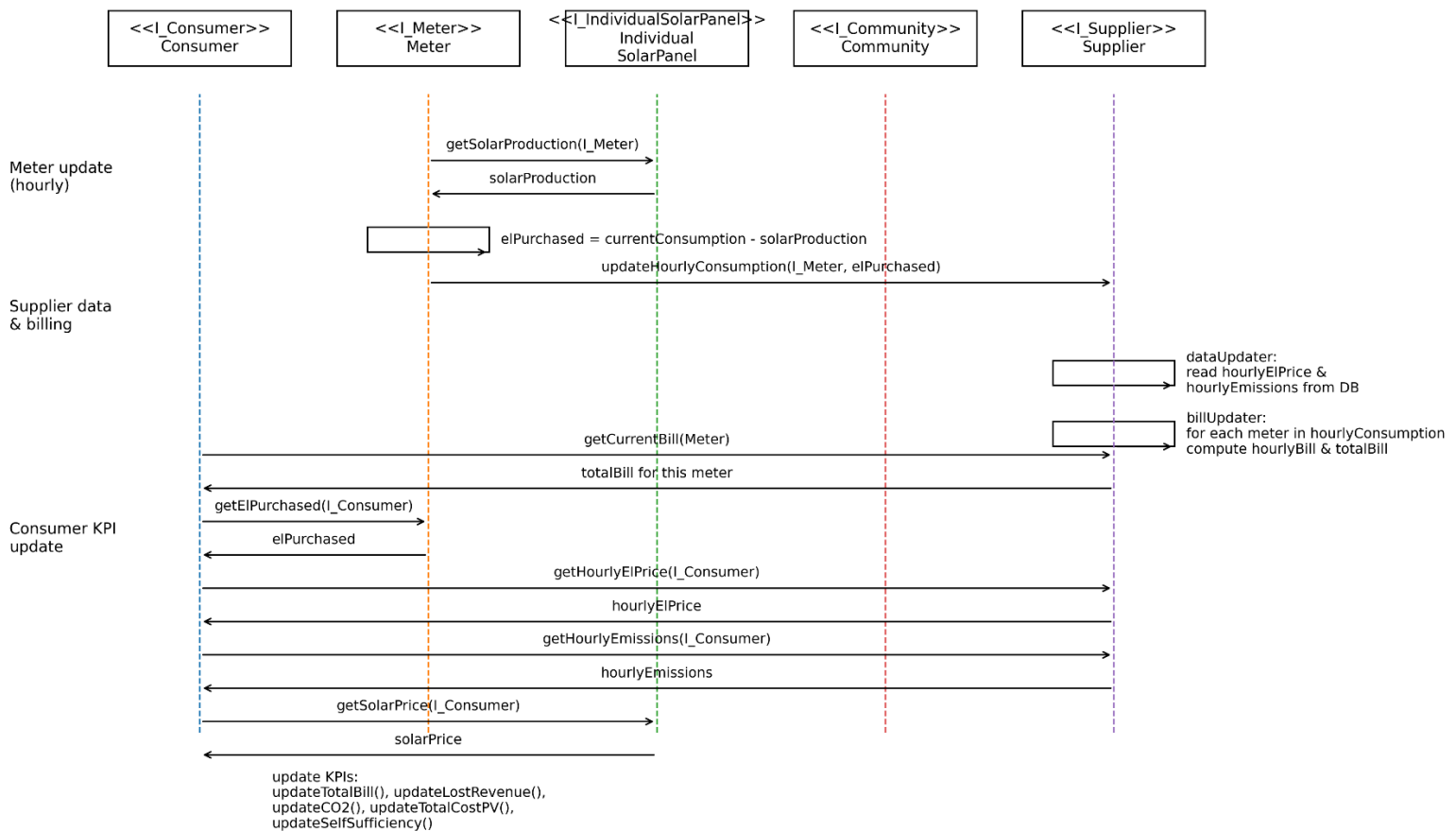


Figure 5: UML Sequence Diagram for Base Case.

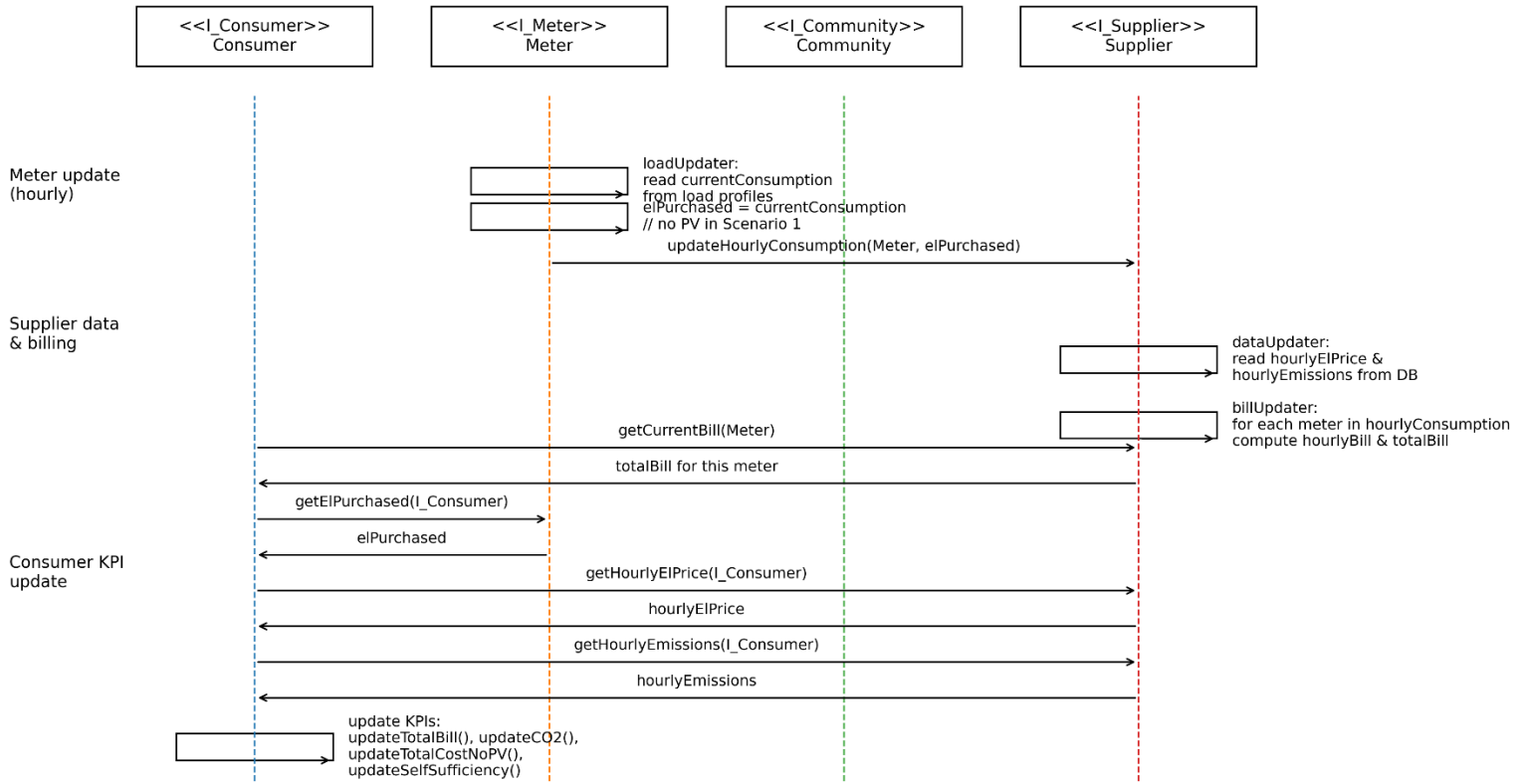


Figure 6: UML Sequence Diagram for Scenario 1.

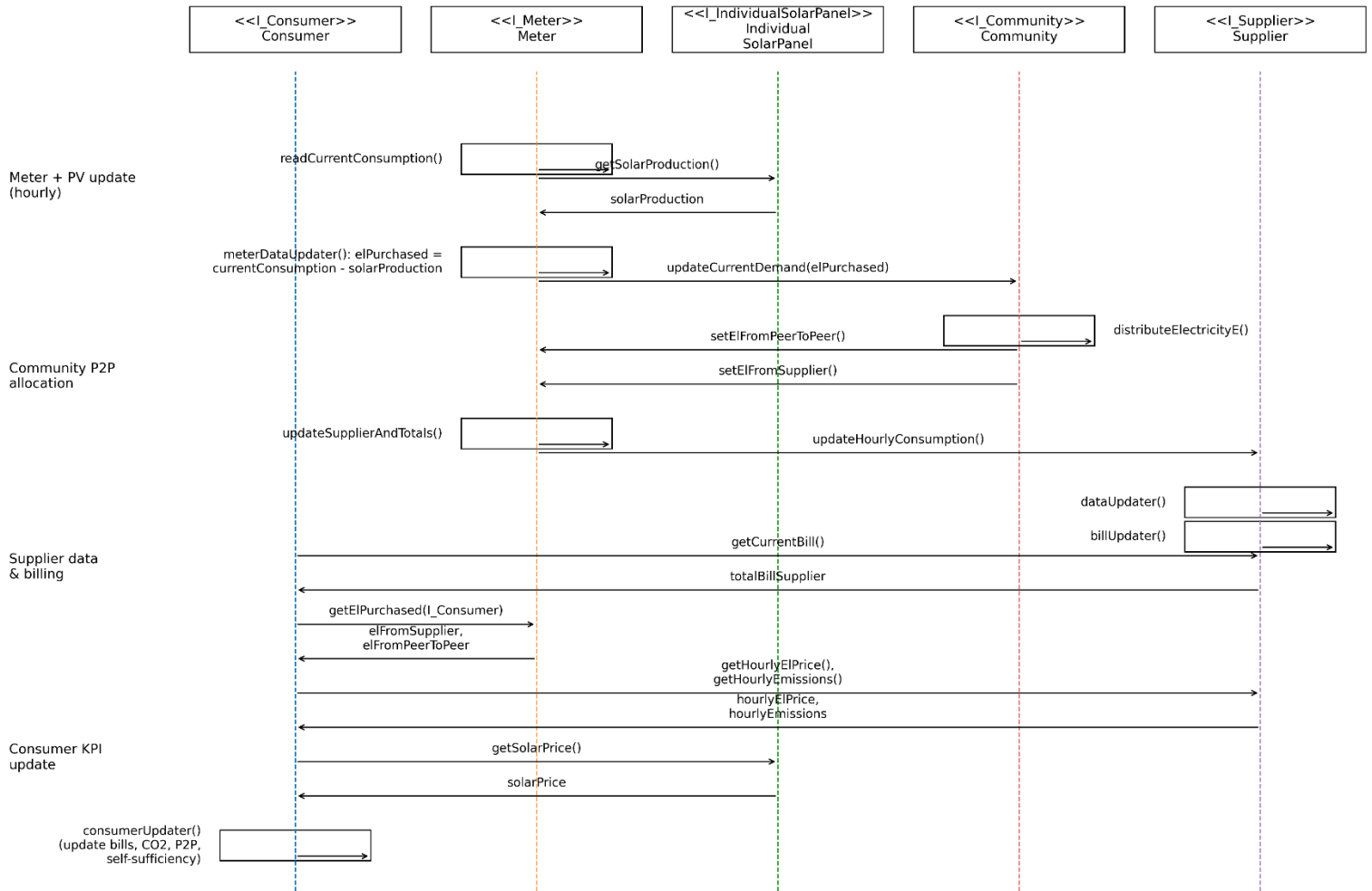


Figure 7: UML Sequence Diagram for Scenario 2 (also representative of Scenario 3, which diagram can be found in Appendix B).

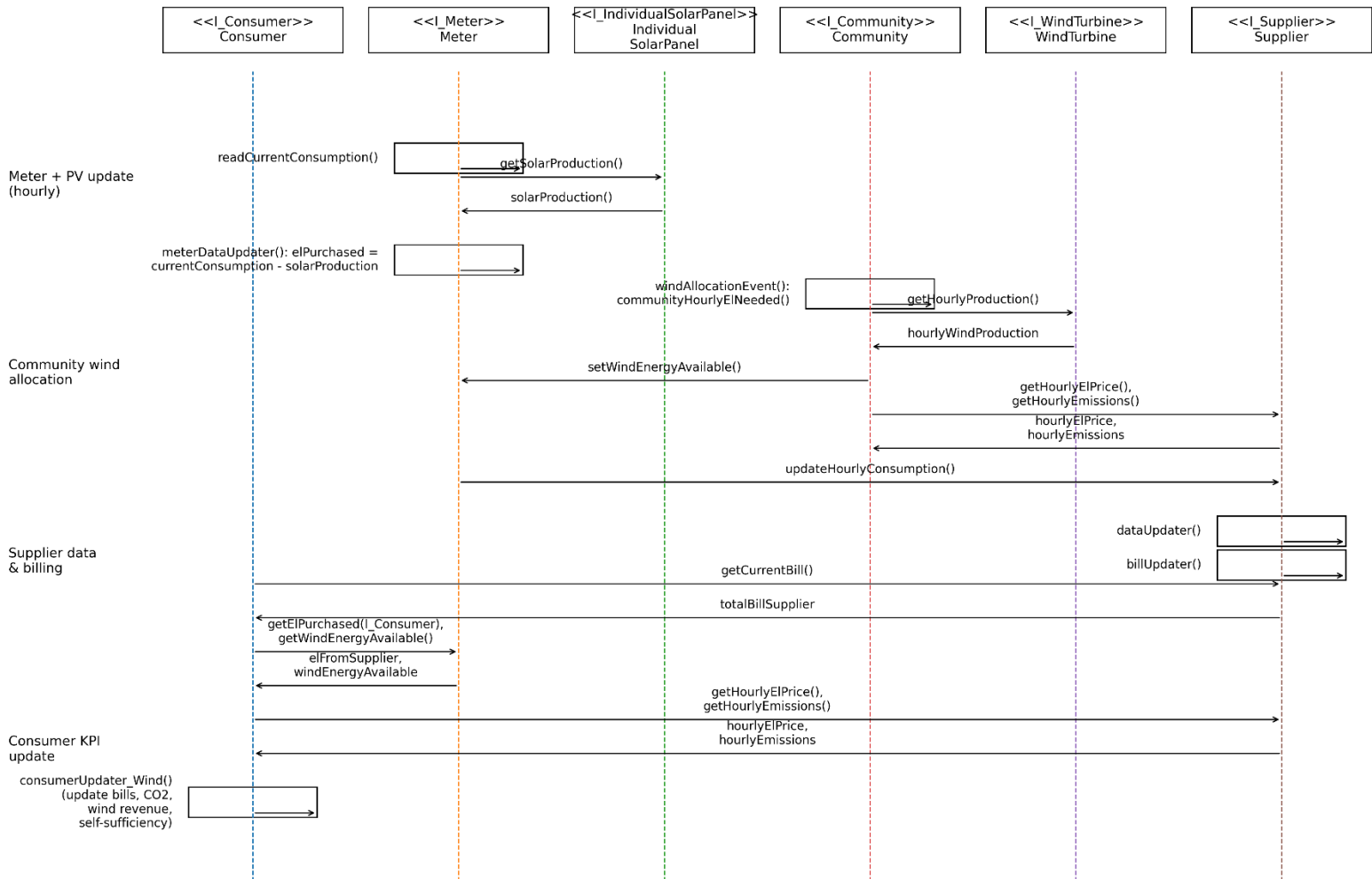


Figure 8: UML Sequence Diagram for Scenario 4 (also -partly- representative of Scenario 5, which diagram can be found in Appendix B).

2.5 Data Input

The table below together with the text after table, provides an overview of the data sources used for the model.

Table 22. Overview of the data sources used in the model

Agent	Data	Using	Source
Meter	Consumption data	Historical electricity consumption data for 137 consumers over a	Dataset <i>2years_137consumerDa</i>

		one-year period is utilized to simulate realistic usage patterns.	<i>taset.xlsx</i> provided as part of course materials
IndividualSolarPanel	Consumption data	Average yearly electricity consumption data is used for dimensioning solar panels.	Dataset <i>2years_137consumerDataset.xlsx</i> provided as part of course materials
Supplier	Electricity spot prices	Is used for calculating the electricity cost and revenue for consumers.	(Open Power System Data, 2020)
IndividualSolarPanel	Historical solar production	Is used together with PV capacity for calculating hourly efficiency and through that hourly production of solar energy.	(Open Power System Data, 2020)
WindTurbine	Historical onshore wind production	Is used together with wind turbine capacity for calculating hourly efficiency and through that hourly production of community wind energy.	(Open Power System Data, 2020)
Supplier	CO2 emissions	Is used for calculating community's carbon footprint together with hourly consumption.	(Energinet, 2024)
IndividualSolarPanel	Technical lifetime and investment data	Is used for calculating the cost of owning a solar panel.	(Energistyrelsen, 2025)
WindTurbine	Technical lifetime and investment data	Is used for calculating the cost of owning a community wind turbine.	(Energistyrelsen, 2025)

A dataset *2years_137consumerDataset.xlsx*, containing hourly electricity consumption for 137 households for years 2015 and 2016, was provided as part of the course material. This data for year 2015 was used as a direct input into the *Meter* agent (stored in a variable

currentConsumption) for modelling hourly electricity consumption patterns. The same dataset was also used for sizing of solar panels (in the model parameter *pVSize* in the agent *IndividualSolarPanel*) by using the following formula:

$$pVSize [kW] = Yearly\ Average\ Consumption [kWh] \div 1000$$

It is important to note that the *Yearly Average Consumption* was calculated for each consumer separately, meaning that the solar panel sizes used in the model are different for each consumer based on their consumption, thus reflecting the real-world situation.

A dataset by Open Power System Data (Open Power System Data, 2020) was used for modelling hourly electricity prices (variable *hourlyElPrice* in agent *Supplier*), solar energy production and wind energy production. For modelling solar and wind energy production, first respective efficiencies (in the model *solarEfficiency* in *IndividualSolarPanel* agent and *windEfficiency* in *WindTurbine* agent) were calculated as follows:

$$solarEfficiency = \frac{DK_solar_generation_actual}{DK_solar_capacity}$$

$$windEfficiency = \frac{DK_wind_onshore_generation_actual}{DK_wind_onshore_capacity}$$

Based on *solarEfficiency/windEfficiency* and *pVSize/WindTurbine size*, *solarProduction* and *hourlyProduction* were calculated as follows:

$$solarProduction = solarEfficiency \times pVSize$$

$$(wind\ energy)\ hourlyProduction = (windTurbine)\ size \times windEfficiency$$

Historical data about hourly CO₂ emissions for electricity (stored in the *Supplier* agent in variable *hourlyEmissions*) was obtained from Energinet dataset available on Energi Data Service website (Energinet, 2024). This dataset contained information about hourly share of each fuel type used in the electricity production in specific price area together with emissions per kWh. The data was aggregated using Python to obtain average hourly CO₂ emissions [g] per kWh energy used.

A dataset by Energistyrelsen containing information about investment data (Energistyrelsen, 2025) was used for calculating hourly cost for PV and wind turbine, as well as dimensioning the community wind turbine. The solar panels were modelled using technical specifications for residential PV in year 2015. Based on its technical lifetime (30 years) and nominal investment (1.67 MEUR/MW_e), the hourly solar panel cost (variable *solarPrice* in the agent *IndividualSolarPanel*), was calculated as follows:

$$(hourly) \text{ solarPrice} = \frac{pVSize [kW] \times 1.67 \left[\frac{MEUR}{MW_e} \right] \times 7.46 \left[\frac{DKK}{EUR} \right] \times 1000}{30 [years] \times 8760 \left[\frac{hours}{year} \right]}$$

The community wind turbine was modelled using technical specifications for domestic turbines in year 2015. For easy comparison between PV and wind turbines performance, the wind turbine was sized using technical lifetime of 20 years and nominal investment cost of 4.25 MEUR/MW_e as follows:

$$(WindTurbine) \text{ size} = \text{cost of added PVs} \times \frac{\text{lifetime of PV}}{\text{lifetime of wind turbine}} \div \text{cost per MW}$$

$$(WindTurbine) \text{ size}$$

$$= 339 \text{ kW} \times 1.67 \left[\frac{MEUR}{MW_e} \right] \times 1000 \times \frac{20 \text{ years}}{30 \text{ years}} \div 4.25 \left[\frac{MEUR}{MW_e} \right] \div 1000$$

$$= 88.8 \text{ kW} \approx 90 \text{ kW}$$

In the model, hourly wind turbine cost was calculated as follows:

$$hourly \text{ windturbine cost} = \frac{windTurbine \text{ size} [kW] \times 4.25 \left[\frac{MEUR}{MW_e} \right] \times 7.46 \left[\frac{DKK}{EUR} \right] \times 1000}{20 [years] \times 8760 \left[\frac{hours}{year} \right]}$$

3. Scenario design

Firstly, it is important to introduce the parameters which do not change between the baseline and designed scenarios. These parameters are presented in the table below, and they are not

mentioned in the breakdown of each designed scenario in this section, meaning that they are fixed (unchanged) throughout all experiments.

Table 33: Fixed Parameters of the Simulation in all Scenarios

Parameters	Settings
Number of domestic consumers	137
Simulation start date and time	01/01/2020 00:00
Simulation end date and time	31/12/2020 23:00
Domestic consumer hourly demand	From dataset
Number of electricity suppliers	1
Total grid capacity (Transformer capacity)	400 kVA
Weather conditions	From dataset
Price of electricity sold from main supplier	From dataset
CO2 Emissions (per kWh)	From dataset

There are also some parameters that are fixed but not always present, like the ones above. There are some scenarios, which will be introduced below, that have PV panels and Wind Turbine (WT), which is accompanied by their cost, their electricity production, their capacities etc. These parameters are listed in Table 5.

Table 44: Fixed Parameters of the Simulation in the Scenarios which they appear.

Parameters (when present)	Settings
Price of surplus electricity sold to main supplier from PV panel owners	20% of the fixed electricity price
Price of surplus electricity sold to peers from PV panel owners	50% of the fixed electricity price
WT capacity (maximum)	90 kWh

The scenarios designed for the purpose of this report is based on real examples of Energy Communities around the world. As mentioned earlier, the baseline scenario (referred as base case in this report) is representative of a ‘normal’ Danish community found in DK1. It is important to mention that, in the case of excess electricity production, the surplus electricity is sold back to the main supplier through the grid. Excess electricity exists when the households with PV panels produce more electricity than they consume in a certain hour, as the production and consumption in the simulations are calculated hourly. In order to test whether the adoption of

PV panels by approximately 20% of the residence is beneficial, both financially and environmentally, it is only logical to compare it with the lack of them; hence the first scenario is created. In scenario 1 it is assumed that the entire community, all 137 households, rely solely on the electricity grid for their electricity supply, and none of them have PV panels. In the case of excess electricity in base case, the price it is sold to the main supplier is 20% of the fixed electricity price.

Similarly, as mentioned in the Introduction, the concept of Energy Communities relies heavily on peer-to-peer (P2P) electricity trading. Essentially, P2P electricity trading allows individuals (in this case, the households) to buy and sell electricity directly from each other, bypassing the traditional grid. Scenario 2 is based on P2P electricity trading within the community, assuming the same PV adoption rate as the base case, 20% of all households. In principle, those 27 households prioritize their electricity consumption, but in the case of excess (also mentioned surplus) production, they sell to their neighbouring households first. In case all households have covered their electricity demand, and there is still surplus electricity produced from the PV panels, then it is sold to the electricity supplier through the main grid. The price the peers pay for the electricity produced by their peers is 50% the fixed electricity price set by the main supplier, and once again, in case of excess electricity sold to the grid, this price is 20% the fixed electricity price.

Scenario 3 is very similar to scenario 2, except for the PV panel adoption level; in this case, all 137 households have PV panels installed. Again, there is P2P electricity trading, with the same rules as in scenario 2: surplus electricity is firstly sold to the neighbouring households, and in case there is still excess after all demand has been met, then it is sold to the main electricity supplier through the grid. This scenario is developed in order to test whether it would be financially beneficial for the entire Energy Community to have PV panels installed, or if Scenario 2 is better, where the community relies on fewer PV panels than households. In both scenarios and in the base case, as well as in the following scenarios, the cost of PV panels (and later wind turbines) is included and considered in the simulation.

After reviewing examples of Energy Communities within Denmark and abroad, it became clear that another very promising approach is a community owned Wind Turbine (WT). Unlike PV

panels, which are privately owned, the WT would be owned by the entire community, and all residents would benefit from the power production. In this case, the excess power produced by the WT would be sold through the grid to the main electricity supplier. The households with individual PV panels, firstly, consume the electricity produced by the panels, and if the demand is not covered, then they consume power produced by the WT. The surplus electricity of the individual PV panels is sold back to the main supplier through the grid, no matter the electricity prices, but in the case of the WT, the surplus electricity is sold to the main supplier only when there are positive electricity prices.

Lastly, as mentioned above, Scenario 5 is very similar to Scenario 4, except for the PV panel adoption level; in Scenario 5, all 137 households have PV panels on top of the community WT. In this case, the individual consumer's demand for electricity is met by the solar power from the PV panels and the community WT. Similarly to Scenario 4, the households prioritise electricity consumption from the PV panels, then from the WT, and lastly from the grid. The surplus electricity of the individual PV panels is sold back to the main supplier through the grid, no matter the electricity prices, but in the case of the WT, the surplus electricity is sold to the main supplier only when there are positive electricity prices.

Table 55: Summary of Scenarios and the Key objects of the simulation

Scenario	Grid Connection	PV Panels	Wind Turbine	P2P
Base Case	✓	✓		
Scenario 1	✓			
Scenario 2	✓	✓		✓
Scenario 3	✓	✓		✓
Scenario 4	✓	✓	✓	
Scenario 5	✓	✓	✓	

The scenarios are designed to represent different stages of a local energy ecosystem transition from a centralized, incumbent configuration toward a coordinated, decentralized, low-carbon system. The ecosystem stages have been analysed in the Introduction, therefore the assignment of the scenarios to them is simple and it follows a similar structure:

1. *Stage 1: 'Incumbent' stage.* As mentioned, all households are connected to the main grid and there are no other sources of electricity present. There is only one scenario that fits in this stage and that is Scenario 1.
2. *Stage 2: Niche Technology Introduction.* In this stage more advanced technologies than the main grid connection is present, like PV panels, but they are not widely adopted. At this stage, the households which have PV panels are also just connected to the main supplier through the grid, and they 'sell' their surplus electricity back to the main supplier. There is no interconnection between the households, and those with PV panels are solely benefitting from them. The scenario fitting in this stage is the Base Case.
3. *Stage 3: Niche Technology Adoption.* In this stage, there are more technologies than just the main grid connection; either PV panels and P2P electricity sharing or community WT. It is important to note that Energy Communities are present only at this stage, as essentially, they are based on community electricity sharing. The community electricity sharing is achieved either through P2P between PV panel owners and their peers, or by being connected to the community WT and also the grid. In both cases, the households are connected to both the main electricity grid and to their peers; this is the state-of-the-art technology adopted in Energy Communities. The scenarios that are in this stage are Scenario 2, Scenario 3, Scenario 4 and Scenario 5.

Overall, the scenarios are built in order to test whether Energy Communities can benefit from P2P electricity trading financially and whether or not it is overall more environmentally friendly than other solutions, like 100% reliance on grid electricity. The Key Performance Indicators (KPIs) in this report represent the main research question; which case is more financially beneficial and environmentally friendly? In order to be able to compare the results of the scenarios regarding this question, the KPIs were developed: regarding the financial aspect, the results are shown in reduced costs (%). The latter represents the amount of money "saved" by not paying the fixed electricity price per kWh of electricity bought from the main electricity supplier, by satisfying some or all of the electricity demand from PV panels or WT. Regarding the environmental aspect, the scenarios are compared based on their CO₂ emissions, calculated in t. The lower the emissions of a scenario, the more ideal it would be for a real-world application in regard to the environment,

while the highest in the financial KPIs would be ideal for the consumers and their financial situation. The last KPI used is self-sufficiency which measures the percentage of community's electricity demand that is created within community and is calculated as follows:

$$\frac{1}{8760} \times \sum_{h=1}^{8760} \frac{\text{electricity covered by community production (excl. export to grid) [kWh]}}{\text{community electricity demand [kWh]}}$$

It is important to note, that hourly self-sufficiency is capped at 1.

4. Result Analysis & Sensitivity

4.1 Result Analysis

Three different models were created to simulate the six scenarios to ensure higher clarity. In all the simulation files, the only parameter that must be manually changed is *householdsWithPV* located in the agent *Community*. The following table provides an overview of simulation files and the parameter values used for each scenario.

Table 6: Overview of simulation files and parameter values.

Scenario	AnyLogic File	Value of the Parameter <i>householdsWithPV</i>
Base Case	Basecase+Sc1_Final_101225	27
Scenario 1	Basecase+Sc1_Final_101225	0
Scenario 2	Sc2+Sc3_Final_121225	27
Scenario 3	Sc2+Sc3_Final_121225	137
Scenario 4	Sc4+Sc5_Final_121225	27
Scenario 5	Sc4+Sc5_Final_121225	137

Figures 9 and 10 together with Table 7 provide an overview of the results. The scenario with highest carbon footprint is scenario 1 (no renewables) with the CO₂ emissions of ~157 tons. While base case, scenario 2 and scenario 4 have also positive emissions, the emissions from scenario 3 and scenario 5 are negative, being respectively ~ -11 tons and ~ -76 tons.

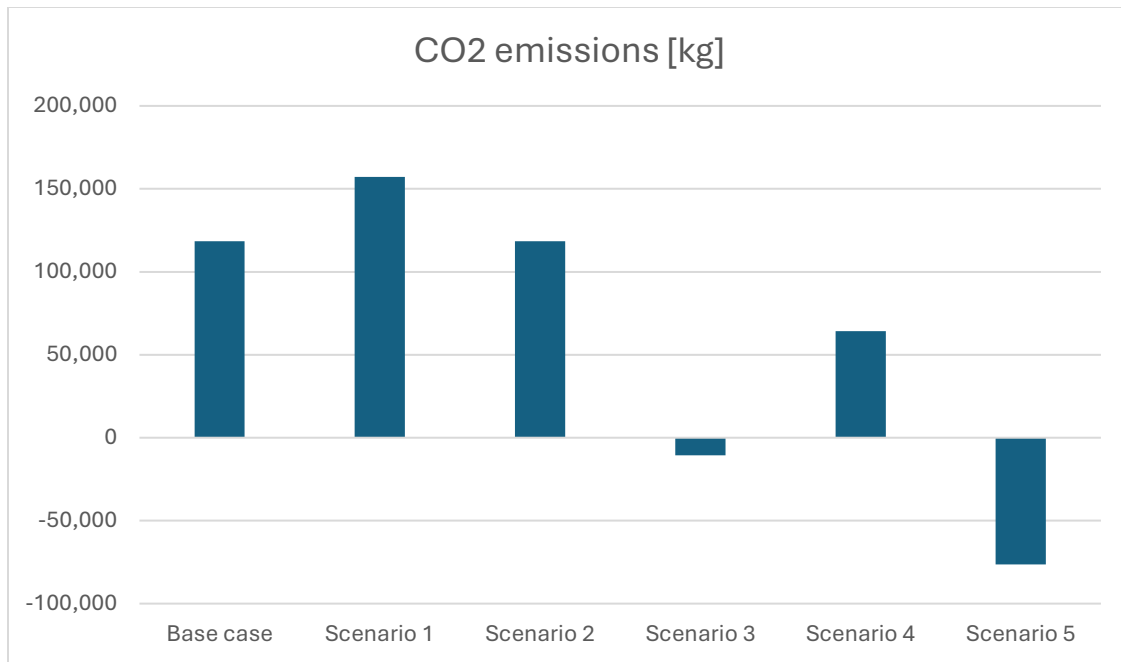


Figure 9. CO2 emissions of different scenarios [kg].

The emissions for the base case and scenario 2 are almost identical (~ 118 tons in both cases) due to the analysis on the community level. While the opposite value of hourly emissions (i.e., the amount of electricity sold to the grid is multiplied by the hourly emissions and -1) is given for electricity sold to the grid, its effect is nullified by the emissions of electricity bought from the grid by other consumers in the community. Similarly, in the case of peer-to-peer sharing, the CO₂ emissions for electricity sold within the community are assumed to be 0, since it comes from renewable sources.

It is noteworthy that the adoption of solar panels by 19.7% of people in the community leads to emission reduction of 24.7% in comparison to scenario 1, and the adoption of solar panels by 100% of people in the community leads to a carbon-negative community when combined with either P2P sharing or a WT.

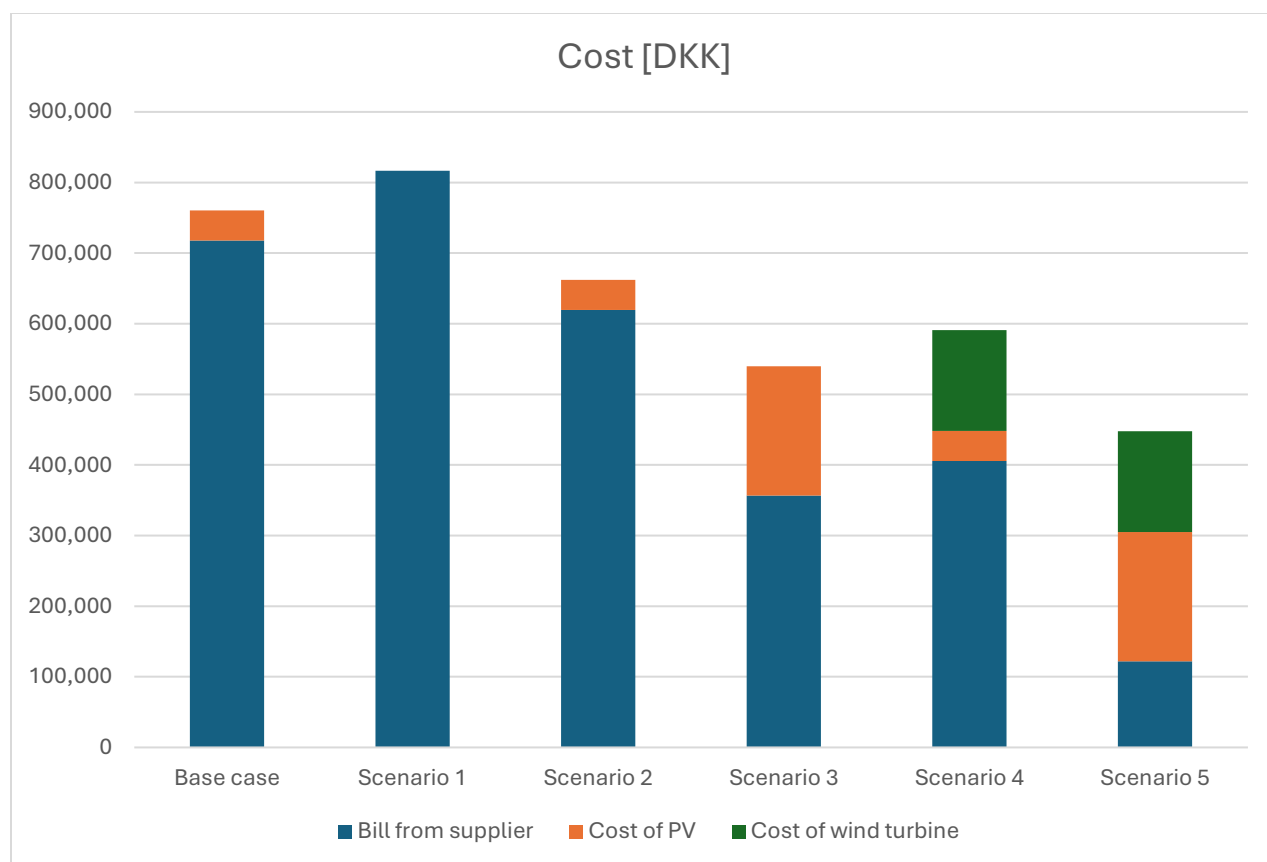


Figure 10. Cost comparison of different scenarios [DKK].

The total cost, consisting of the supplier bill (from which revenue from selling electricity to the grid has been deducted), cost of PV and wind turbine, is highest for scenario 1 (816,835 DKK) and lowest for scenario 5 (447,765 DKK), indicating that the most environmentally friendly solution is also the cheapest. However, this does not account for any emissions originating from the production of wind turbines or solar panels.

While base case and scenario two had almost identical emissions, the cost of scenario 2 is approximately 100,000 DKK or 12.9% lower than the cost of base case, demonstrating clear benefits of P2P energy sharing.

Scenario four (27 households with PV + WT) performs worse in terms of both CO₂ emissions and costs than scenario three (137 households with PV), despite Denmark's high wind potential. This can potentially be explained by low electricity prices during the hours of high wind energy production due to Denmark's large wind capacity, thus not leading to expected savings.

Table 7: Overview of simulations results.

Scenario	Bill from supplier [DKK]	Cost of PV [DKK]	Cost of wind turbine [DKK]	CO2 emissions [kg]
Base case	717,929.97	42,357.88	-	118,388.99
Scenario 1	816,834.58	-	-	157,208.03
Scenario 2	619,590.90	42,357.88	-	118,426.62
Scenario 3	356,645.98	183,135.54	-	-10,611.20
Scenario 4	405,797.79	42,357.88	142,672.50	64,316.09
Scenario 5	121,956.72	183,135.54	142,672.50	-76,363.17

In scenario 2 (27 households with solar panels + peer-to-peer sharing), 72.98% (332,115 kWh out of 455,080 kWh) of electricity when measured as yearly total (i.e. not accounting for hourly fluctuations in electricity sold to/bought from grid) came from the grid, 14.71% (66,955 kWh) from peer-to-peer sharing and 12.31% (56,009 kWh) from own solar panels (see Figure 11). In scenario 3 (137 households with solar panels + peer-to-peer sharing), only around 1.00% of

energy (measured as yearly total) came from peer-to-peer sharing with rest coming from own solar panels.

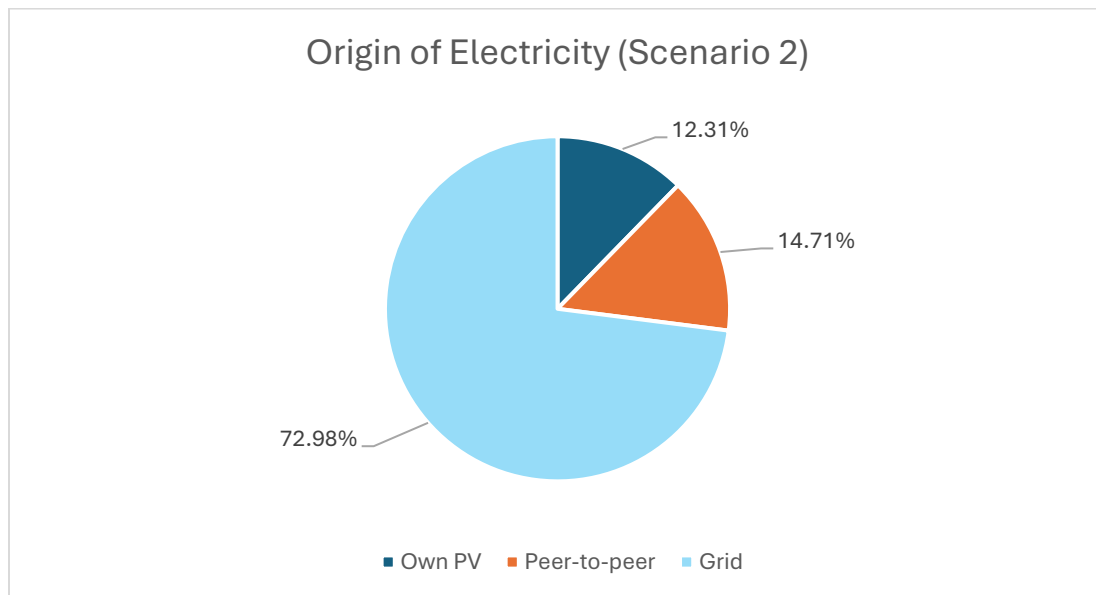


Figure 11. Origin of electricity by source (scenario 2).

Figure 11 shows the yearly potential electricity production for wind and solar and the contributions to self-sufficiency. The yearly total potential for solar energy production is ~123 MWh for the installation capacity of 102 kW (corresponds to 27 households having solar panels) and ~532 MWh for the installation capacity of 441kW (corresponds to 137 households having solar panels). The total potential for wind energy production for the installation capacity of 90 kW is ~230 MWh. This means that at ~12% lower capacity, a wind turbine produces 1.87 times as much energy as solar panels. However, WT benefits more economically by scaling the capacity, and as a direct consequence of a small community, the WT is ~2.54 times as expensive as PV per capacity.

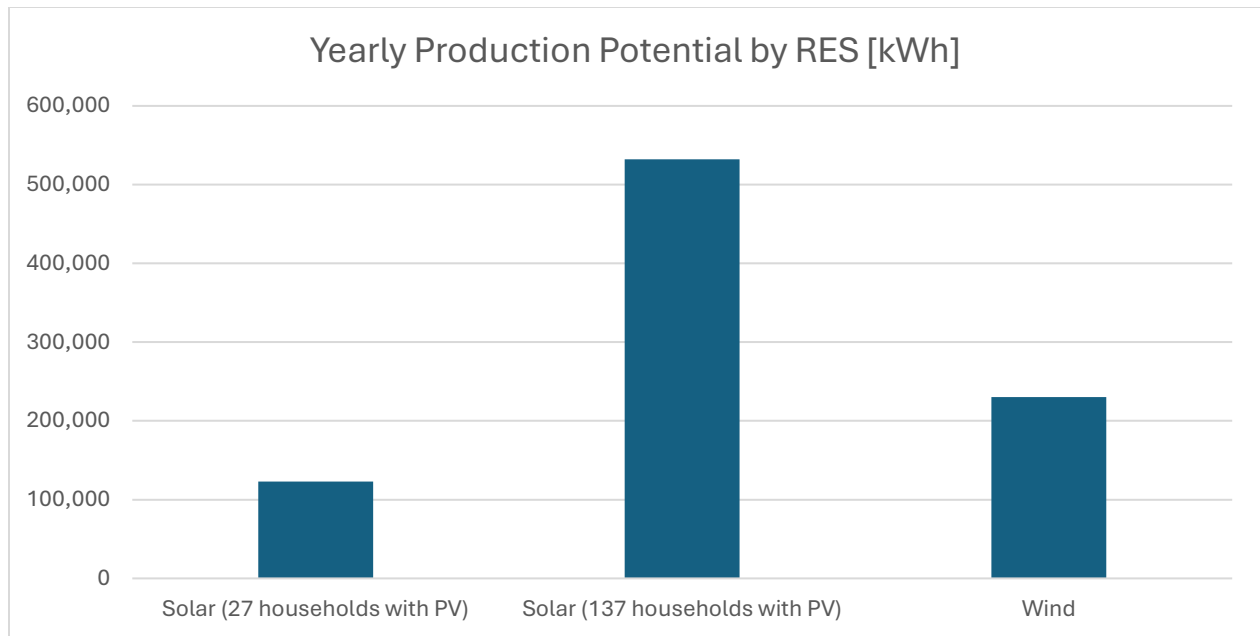


Figure 12. Yearly production potential of different renewable energy sources in kWh (installation capacities from left to right: 102 kW, 441 kW and 90 kW).

4.2 Sensitivity Analysis

Additionally, a sensitivity analysis was conducted to assess the simulation's resilience to potential change. The parameter chosen was the productivity of the solar panels, specifically $\pm 20\%$ of solar production. Respectively, this should directly increase or decrease the attractiveness of the PVs.

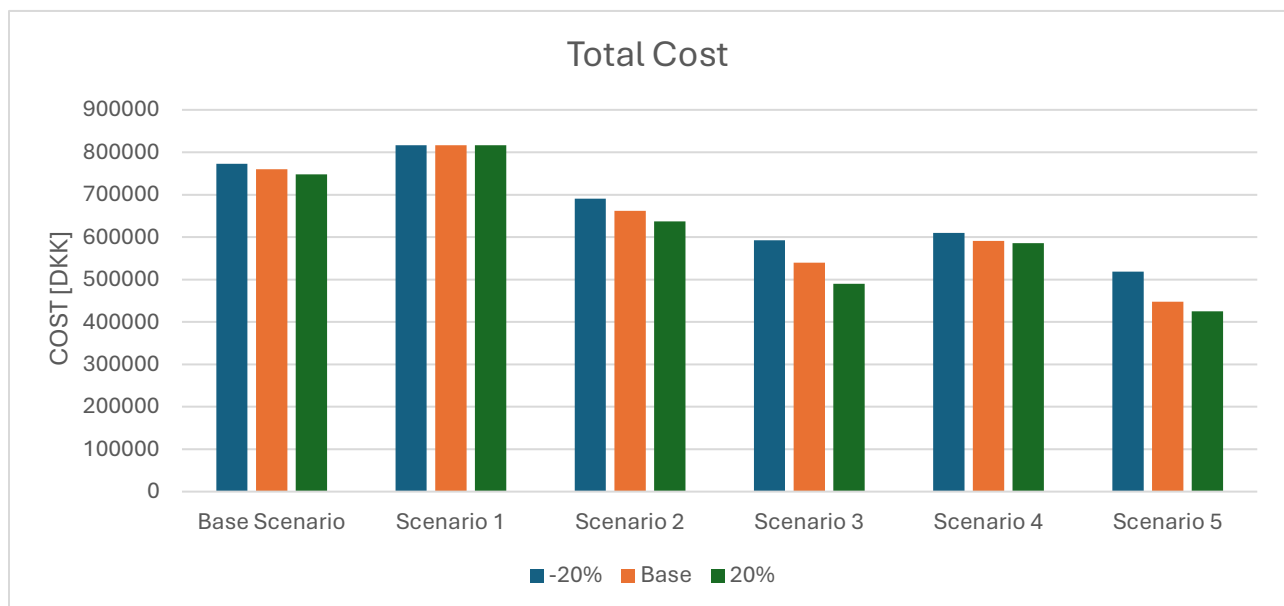


Figure 13: Cost Sensitivity

The sensitivity simulations suggest that having no PVs and importing everything from the grid (Scenario 1) is clearly the most expensive solution, even with the addition of increased parameters. The figure also highlights that scenario 5 is still the cheapest option, with scenario 3 in 2nd place.

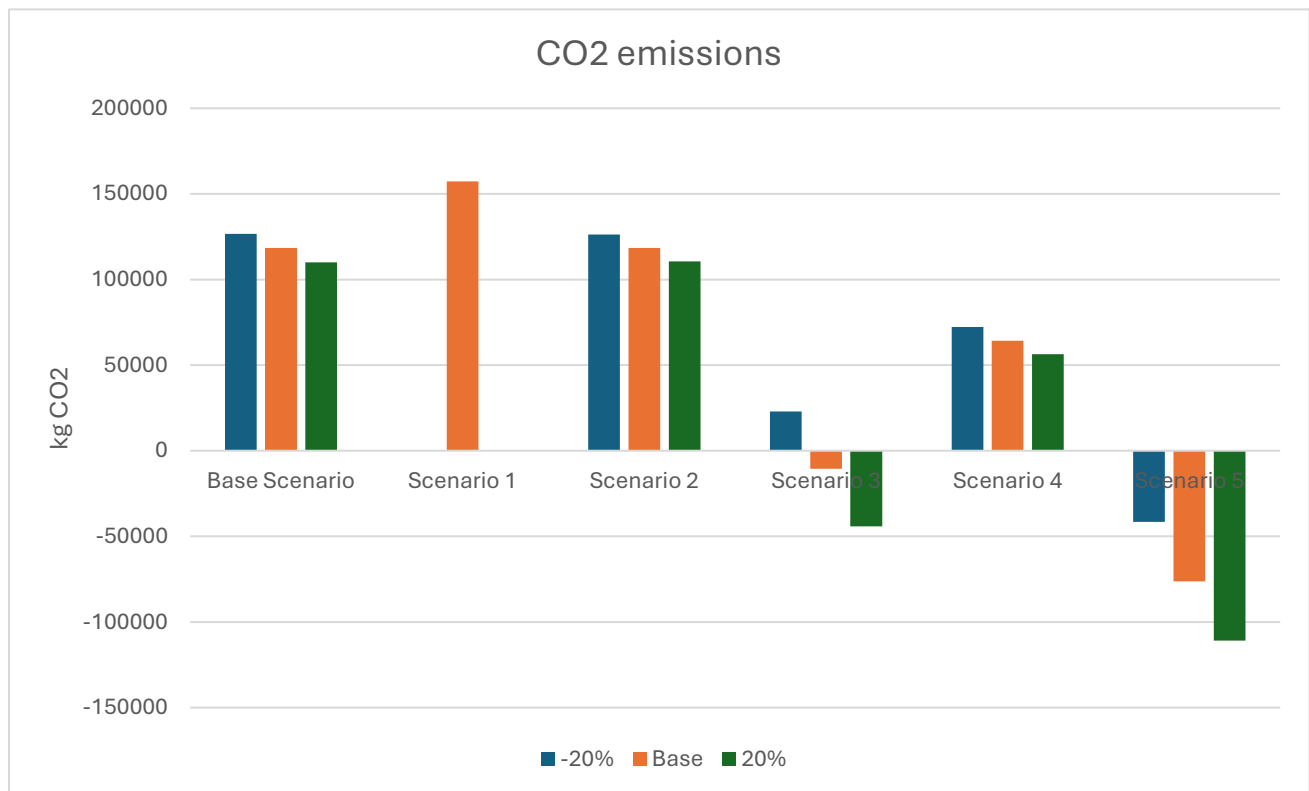


Figure 14: CO2 emissions Sensitivity

The results on the emissions side for the sensitivity still indicate that the base scenario and scenario 2 have identical emission impacts. However, it also highlights the importance of a sensitivity analysis, as scenario 3 can shift to a positive emission impact. Scenario 5 is also the best-performing scenario in this topic, but this could be explained by the additional amount of energy produced.

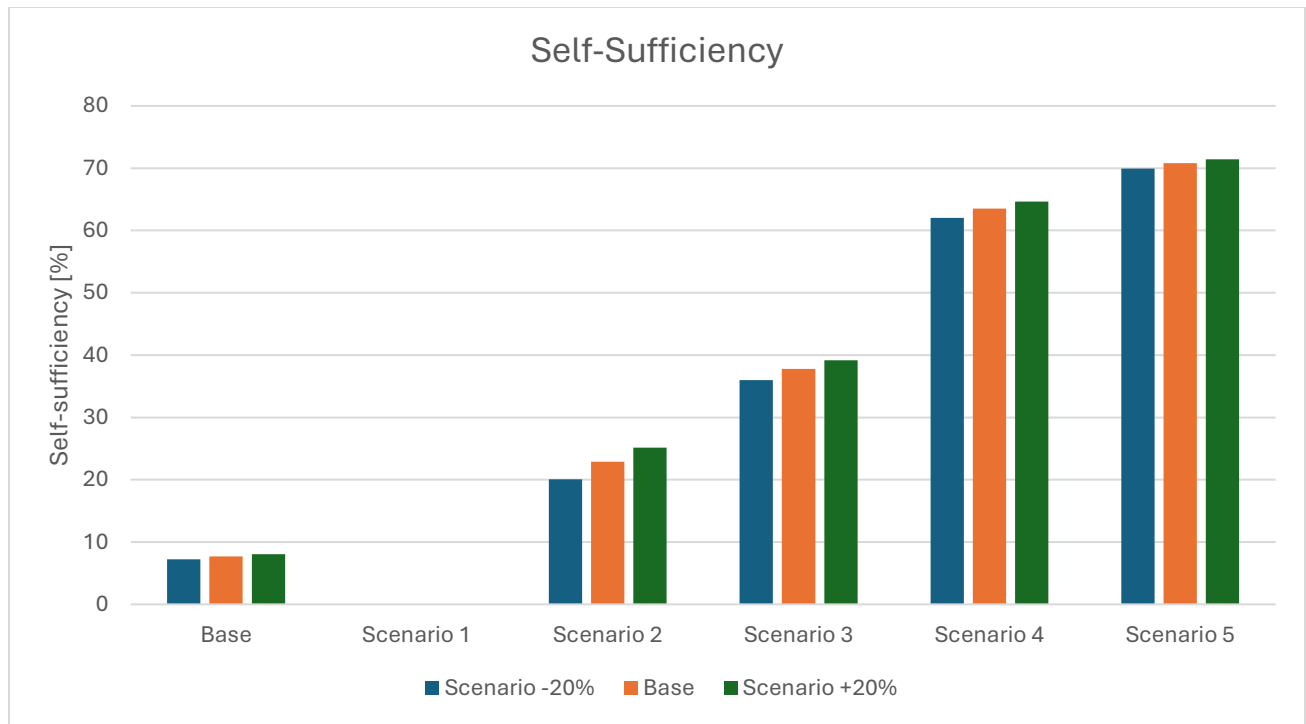


Figure 15: Self-Sufficiency Sensitivity

Expanding upon Figure 11, the self-sufficiency is seemingly improved in scenarios with P2P, with further gains from the addition of expanded PV capacity and additionally improvements from the adoption of WT capacity. Noticeably, all self-sufficiencies only marginally change in comparison to the varying production, which indicates a high resilience. More detailed data for the sensitivity section can be found in the Appendix C - Results and Calculations.

5. Discussion

5.1 Experiment testing results

The study was mainly focused on four different research objectives, which would lead us to an answer to the correlating research question. Additionally, each specific research objective will be discussed and answered in this section based on the results and simulations.

The first research objective is focused on introducing a more realistic consumption and production of each household through the use of individual demand and incorporated exogenous factors. To answer the research objective, the simulations were run on an hourly basis and on an

individual level, meaning that every single household (137) had their own hourly demand, and different PV production based on sizes in correlation to their household demand. This was further showcased by the use of several scenarios as some scenarios incorporated 27 households with PV and others with all 137 households.

The simulations contained data for an entire year, and the PV production was normalized to reflect real-world PVs. Therefore, if they were affected by bad weather, no irradiation, or season variations so would the simulated PVs. As expected, the PVs did not produce during night and had increased production in summertime. This objective was conducted on all scenarios to simulate a realistic profile, which was extended upon with other objectives.

The second research objective, set the stage for using peer-to-peer and enabling an energy community, that benefits everyone economically (and environmentally). The current situation is that a household with excess amount of production is forced to sell it to the grid for approximately 20% of the spot price. However, enabling a P2P strategy with a fixed rate of 50% based of the varying spot prices, allows the community to create an attractive economic incentive for both parties. In short, selling the electricity for 50% (instead of 20% to the grid) by enabling P2P means the consumer pays 50% of the spot price when excess amount of electricity is available, while the producer is paid an additional 30%. Saving the community 80% of the total economic impact by using P2P instead of exporting to the grid. Additionally, it is expected that this market strategy will create an economic incentive to attract more PV investments in the future. This would continue until the community gradually meets an economic equilibrium, where additional PVs do not provide sufficient economic returns. Consequently, other implemented P2P strategies simply distribute the benefits differently at an individual level but provide the same value at the community level.

To test the P2P strategy, different scenarios was designed with different parameters. The simulation suggests that the addition of P2P (scenario 2) would reduce the bill for the community by 98,340 DKK or 13% in comparison to the base case (without P2P). However, using P2P did not have an impact on the CO₂ emissions as the base case would simply export renewable electricity

to the grid which nullifies the impacts from the imported electricity to the community, hence also operating and performing alike P2P.

The remaining scenarios further examine a variety of options for the economic and environmental impacts by increasing the number of PVs and adding WT capacity. The 3rd scenario is an extension of scenario 2 but has increased the number of PVs from 27 to 137 households resulting in a reduced bill of 220,506 DKK or 29% in comparison with the base case. Additionally, increasing the amount of PV reduces CO₂ emissions from 157,707 kg to -10,137 kg. Moreover, the WT scenario 4 and 5 highlights the benefit of further increasing the production by reducing the total cost by 39%. However, this additional 10% reduction from scenario 3 to 5 is the product of doubling the capital investment, which potentially could be spend more efficient in other parts of the community.

The third research objective examined the deployment of PV and WTs, and the self-sufficiency of the community. For this research objective, both PV and WT was implemented with different profiles and efficiencies. In addition, a community-level energy and self-sufficiency table was conducted from Figure 12 and Figure 15.

Table 8: Self-Sufficiency

Scenario	Community-level energy [kWh]	Community self-sufficiency [%]
Base	123,080	7.7%
Scenario 1	0	0%
Scenario 2	123,080	22.9%
Scenario 3	532,139	37.8%
Scenario 4	353,193	63.5%
Scenario 5	762,253	70.8%

It should be noted that the self-sufficiency is based on the hourly self-sufficiency over the whole year e.g. the base case could maximum reach 20% as it is only able to supply 20% of the community each hour. Also, all communities consume the same amount of energy, but differs in production, which is what the study refers to as community-level energy.

The fourth research objective investigate the resilience of the scenarios through a sensitivity analysis on the PV production with a variation of $\pm 20\%$, the study focuses on a combination of economic robustness and self-sufficiency. The sensitivity results indicate that the base case is impacted symmetrically by disturbance as it mainly relies on imports from the grid and exports all excess amounts resulting in an approximately variation of $\pm 12,644$ DKK or 1.7%, this indicates a high dependence on the grid which indirectly creates a high resilience of varying PV production but has the disadvantage of a low self-sufficiency as mentioned in a previous objective.

Scenario 2 uses the same setup but has implemented P2P resulting in slightly shifted results of roughly +28,954 DKK or -25,212 DKK respectively +4.4% or -3.8% variation, clearly showcasing the P2P system varies more based on the solar availability, although always outperforming the base scenario. However, scenario 2 (P2P) highlights an asymmetrical economic performance, that reduces the returns more than it improves it, when production varies by $\pm 20\%$. This indicates that P2P incrementally reduces the economic return of the energy but still provides a positive value at all points. Other than the economic benefits of P2P, it also seemingly increases self-sufficiency at all variations.

For higher capacities, such as the +20% case in Scenario 5 (without P2P) the system approaches a saturated limit where renewable generation (wind and solar) often exceeds the individual household demand for everyone, which results in much excess energy being exported. However, this changes with variation, as the return from 20% more production is only 5.0% whereas the reduction from -20% production is a staggering negative 15.7% return, indicating that more production is essentially futile, but less production is unfavourable with high capacity and no P2P. Essentially, making the potential addition of P2P favourable in the case of negative variation. Scenario 3 with P2P on the other hand seems to handle this problem better, increasing/decreasing the cost by around 9.8% or -9.2%, this difference can be appointed to the P2P. For the rest of the scenarios, the specific sensitivity details are shown in the table below.

Table 9: Resilience and sensitivity of the Scenarios

Scenario	Configuration	Disturbance response (Economic)	Disturbance response (Self-sufficiency)	Resilience
Base Case	No P2P + PV baseline capacity	Base Value: 760,287 DKK Change: $\pm 1.7\%$ $\pm 12,644$ DKK	Base Value: 676 Hours / 7.72% Change: + 0.35% / -0.48% +31 / -42 Hours	Low economic resilience, low self- sufficiency, high stability Overall: Weak resilience
Scenario 1	Grid-reliant	Scenario Value: 818,723 DKK Change: $\pm 0\%$ 0 DKK	Scenario Value: 0 Hours / 0% Change: $\pm 0\%$ ± 0 Hours	No local economic resilience, no self- sufficiency, dependent on grid.
Scenario 2	P2P + PV baseline capacity	Scenario Value: 661,948 DKK Change: +4.4% / -3.8% +28,956 / -25,212 DKK	Scenario Value: 2,009 Hours / 22.93% Change: +2.23% / -2.85% +195 / -250 Hours	Better economically, self-sufficient and medium-high stability
Scenario 3	P2P + PV increased capacity	Scenario Value: 539,781 DKK Change: +9.8% / -9.2% +53,004 / -49,676 DKK	Scenario Value: 3,312 Hours / 37,81% Change: +1.35% / - 1.80% +118 / -158 Hours	2 nd best performing economically and sustainability, 3 rd best self-sufficiency.
Scenario 4	No P2P + PV baseline capacity + WT capacity	Scenario Value: 590,828 DKK Change: +3.2% / -0.8% +19,003 / -4,789 DKK	Scenario Value: 5,565 Hours / 63,53% Change: +1.13% / -1.45% +99 / -127 Hours	2 nd best performing self-sufficiency, but 3 rd best economically, very stable.
Scenario 5	No P2P + PV increased capacity + WT capacity	Scenario Value: 447,764 DKK Change: +15.7% / -5.0% +70,451 / -22,507 DKK	Scenario Value: 6,204 Hours / 70.82% Change: 0.65% / -0.86% +57 / -75 Hours	Best performing and highest self- sufficiency stability, but lowest economic stability

Hereby, the simulation indicates that a systems economic optimality is achieved by increasing capacity up to a point where grid exports remain limited. As beyond that point, additional capacity yields reduced returns. Additionally, it appears for the same investment PV has a significant higher economic return than WT.

In the case of self-sufficiency, it is shown that P2P helps on an hourly basis but is still unable to load-shift to other hours. It also shown that WT contributes more than PV as seen by the difference of scenario 3 and 4. From the five scenarios a couple of strategies could be identified to contribute to the system, regarding both self-sufficiency and economics.

Firstly, a system should contain a mixture PV and WT to contribute both economically, but also to enhance self-sufficiency by shifting production across more hours.

Secondly, a system should optimize renewable capacity based on the aggregated demand. As over dimensioning reduces economic return as more energy is forced to the grid, notably this over dimensioning is seemingly less penalized with P2P.

Lastly, adding storage would enable needed load-shifting, and excess amount of energy could be used in off-peak hours to save costs and increase self-sufficiency.

Overall, the research objectives indicate that by using P2P the total energy cost of the community can be reduced significantly, but it does not have a decreased environmental impact. However, adding more local renewable generation capacity to the grid seemingly both has a significance for the economic and environmental impacts of the community.

Economically, the adoption of P2P within a community is cheaper than exclusively exporting to the grid. By solely adopting P2P the community is cutting the total cost by ~13% (Scenario 2), or by ~22% (Scenario 3) if it is combined with more capacity. These reductions arise from excess electricity providing more value within the community than sold to the grid, but increasing the capacity incrementally diminish the returns for P2P as the total demand is saturated in more hours and excess is sold to the grid. Furthermore, the usage of additional WT capacity was tested, and the results indicated an additional reduction of ~11% in total cost, but with an increase of around 78% of the initial investments.

Environmentally, the results showed clear benefits of having more renewable generation enabling the community to be net-negative by reducing CO₂ emissions from 157,707 kg to -10,137 kg by simply expanding from 27 households to 137 households, further increased to -76,363 kg by adding WT capacity. Demonstrating that renewable generation can fully offset and even surpass a local community's environmental impact.

The resilience analysis highlight that the system is significantly affected by variation in PV production. It showcases that P2P in low-capacity scenarios (e.g. scenario 2) is still operating at a high economic efficiency, where a variation of -20% results in a cost increase of +4.4%, meanwhile a +20% variation only reduces the cost by -3.8%. Without P2P, and with a high capacity such as scenario 5 it seems to have diminishing returns at only -5.0% for +20% production and -20% production increases the cost by +15.7%, which indicates additional electricity is sent to the grid.

The findings from the research objective confirm that an energy community can achieve significantly lower costs and lower emissions while increasing the self-sufficiency, which answers the research question:

“Are the Energy Communities more financially beneficial for the residents and more environmentally friendly than normal Danish communities?”

5.2 Adoption Potentials in the targeted Ecosystem

The evaluated technologies on this report are individual households PV panels, community WT and P2P energy trading between the aforementioned households. The results showcase that all three technologies have advantages and realistic adoption potential in the (near-)future ecosystem.

First, the detailed modelling of individual households enables communities or municipalities to design solution that reflect actual household patterns instead of aggregated averages. This results in higher accuracies for forecasting and planning of capacity and potential investment decisions e.g. the difference observed in scenario 3 and 5. For the individual PV panels, the model shows that even a partial adoption level (20% of households – Scenario 1) already reduces grid

purchases and CO₂ emissions, while full adoption combined with self-consumption or P2P trading significantly increases community self-sufficiency.

Secondly, the increase of local renewable capacity clearly has an economic benefit as households remove the costs associated with grid tariffs and losses. The addition of a community WT reduces grid demand and the associated CO₂ emissions.

Finally, P2P showcases a strategy to increase the value of renewable capacity. In the model it enables surplus PV (and indirectly wind) to be reallocated within the community instead of being exported cheaply to the grid, which improves local self-consumption and can lower bills for households with no PV panels. While there are a few real-world examples of community WT in Energy Communities, the simulation also shows that, given cutting-edge technologies (such as meters), such schemes could realistically operate in the current or near-future ecosystem. In practice, adoption will depend on regulation and market design in the respective area and/or country.

Overall, the scenarios suggest that a combined adoption of PV, community WT, and P2P trading has strong potential to address the main challenges of the ecosystem as presented above, while remaining technically plausible.

6. Conclusion

The study successfully created an agent-based simulation model that was able to answer the primary research question: *“Are the Energy Communities more financially beneficial for the residents and more environmentally friendly than normal Danish communities?”*. The results indicate that energy communities were able to decrease their CO₂ emissions, meanwhile increasing economic returns and self-sufficiency. Hereunder, it was demonstrated that the 137 households performed best by adopting P2P and integrating a mix of renewable technologies to spread production across more hours. The key results can be put into three categories, which highlights the best outcomes:

- **Economic Benefits:** The adoption of P2P only provides positive economic value within a community as seen by the 13% cost reduction between scenario 2 and the base case.

Furthermore, the most viable solution was scenario 5 with the largest capacity investment lowering the total from 816,834 DKK to 447,765 DKK.

- **Environmental Benefits:** While P2P provides economic benefits, it does not reduce emissions on a national level. The energy communities' emissions are reduced based on renewable generation, which enables high-capacity scenarios (scenario 3 and 5) to shift into net negative emissions, with Scenario 5 (energy mix) achieving -76 tons of CO₂ emissions.
- **Self-Sufficiency:** The results indicated that a combination of wind and solar maximized the self-sufficiency to 70.8% in scenario 5. It is shown that relying on all your capacity on single technology like scenario 3 results in a high dependence to the grid of 37.8% in comparison to 63.5% from scenario 4 which also adopts WT.

The study highlighted several significant insights, which could enforce the study if it were to be extended upon. Hereunder, the analysis highlighted that from the same investment PVs provides higher economic returns than WTs but contributes less to self-sufficiency and resilience. This means for an energy community that want all benefits it would need a mix of renewable energy generation with P2P. While the study was able to indicate beneficial technologies through analysis, for future study an added scenario with all the beneficial technologies could provide interesting results. Further future improvement could be the introduction battery storage for load-shifting potentially increasing our self-sufficiency and reducing grid exports.

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Appendix A – Assumptions

There are some key assumptions which were included in the execution of the simulation and this report, and these are:

- The grid can 'accept' as much electricity as the surplus electricity that is fed into it from the community to the main supplier. Cable capacities are not taken into account.
- There is no electricity storage taken into consideration.
- PV panels operate continuously, and excess energy is sold to the grid even during the hours with negative electricity prices leading to potential monetary loss.
- Wind turbine will only sell excess energy to grid during the hours with positive energy prices (profitable hours).
- CO₂ emissions for the electricity produced by solar panels and wind turbine is 0g per kWh, when used within community and $(-1) * \text{hourly CO}_2 \text{ emissions [g/kWh]}$ when sold to grid.
- The following priority order for distributing solar and wind energy is used:
 - First the households use the energy produced by their solar panels themselves.
 - If there is excess solar energy, then it is first used by other households and only in case of further excess sold to grid.
 - Wind energy is used in the community only when the demand cannot be covered by solar energy.
- The sensitivity analysis does not consider production variabilities effect on grid prices.

Appendix B – UML Sequence Diagrams (Sc. 3 & Sc. 5)

In the following Figures (16 and 17), the UML Sequence Diagrams for Scenarios 3 and 5 respectively are presented. They are very similar to Scenarios 2 and 4 respectively, with mainly numerical differences, which are not displayed in the diagrams.

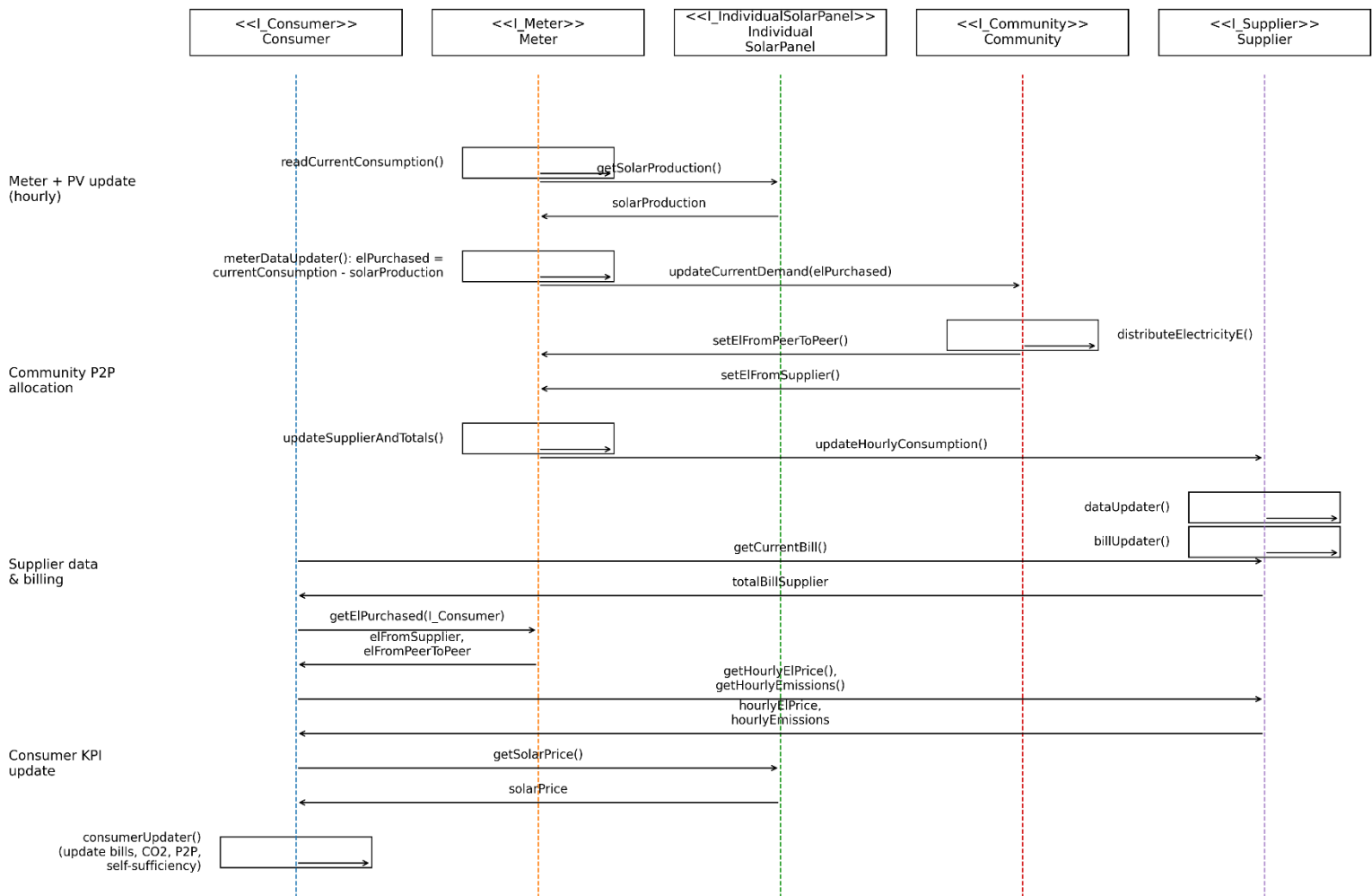


Figure 16: UML Sequence Diagram of Scenario 3

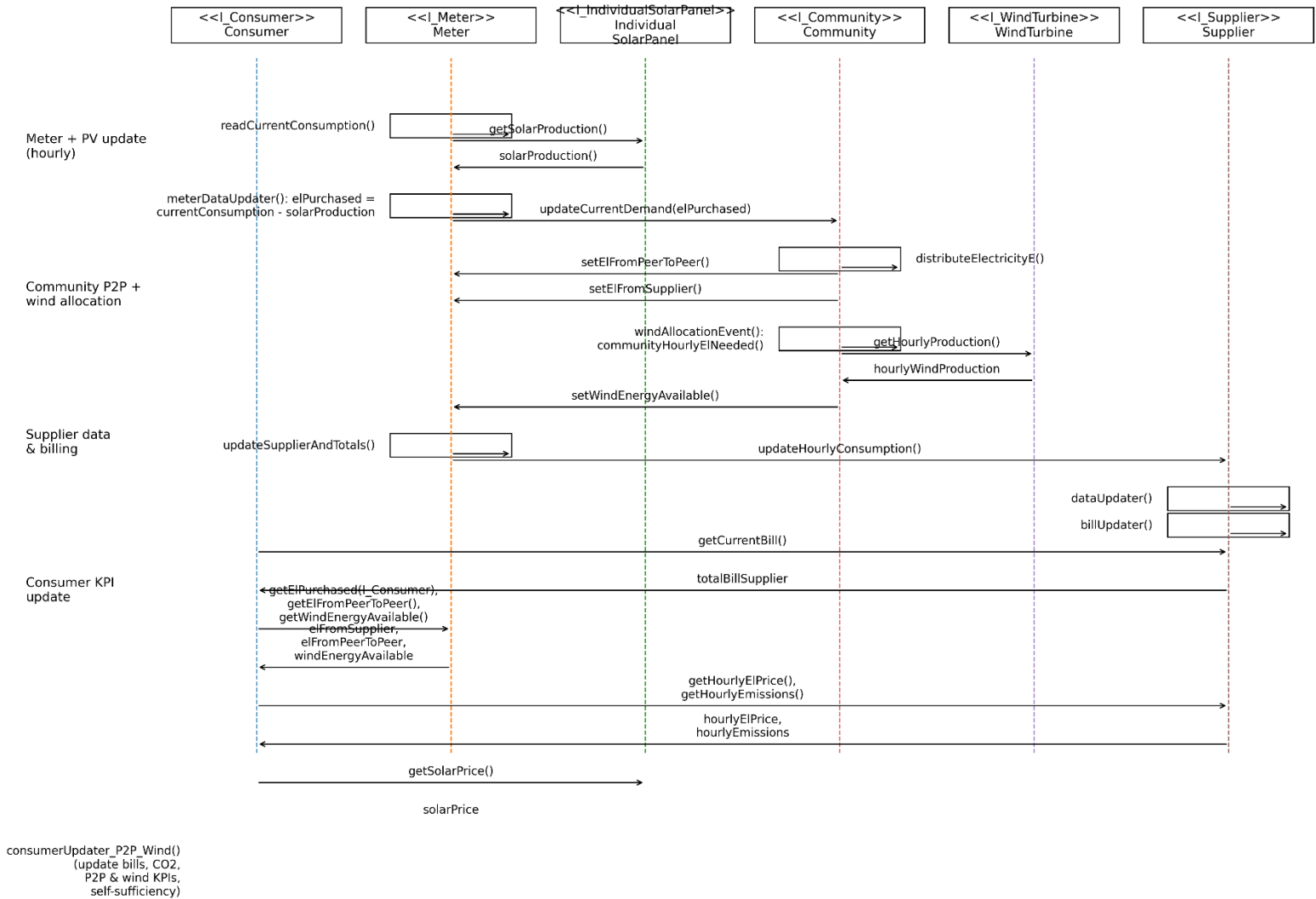


Figure 17: UML Sequence Diagram of Scenario 5

Appendix C - Results and Calculations

This includes the attached excel called “result excel.xlsx” that holds the results, and calculations done in this project.

Appendix D - Simulation, datasets and other

Overall, the study has produced a couple different simulations, and few modifications needs to be made to initiate certain scenarios and their respective sensitivity.

To run the different scenarios, different number of PV is needed as shown in Table 5
To run the sensitivity simulations the solarproduction needs to be modified as shown:

$$\textit{solarProduction} = \textit{solarEfficiency} \times \textit{pVSize} \times \textit{Sensitivity} \textit{ (E.g. 0.8 or 1.2)}$$